International Journal of Geo-Information

MDPI

Article

A Machine Learning Approach Using Spatially Explicit *K*-Nearest Neighbors for House Price Predictions

Meifang Chen, Changho Lee * and Yongwan Chun

School of Economic, Political, and Policy Sciences, The University of Texas at Dallas, Richardson, TX 75080, USA; flychmf@gmail.com (M.C.); ywchun@utdallas.edu (Y.C.)
* Correspondence: changho.lee@utdallas.edu

Abstract

Spatial data has distinctive properties that differentiate it from non-spatial data. One prominent characteristic is spatial autocorrelation (SA). When machine learning techniques are applied for spatial data modeling, they require spatially explicit consideration. If these inherent spatial structures are ignored, models may produce biased predictions. However, integrating this property into the model yields additional spatial insight, thereby enhancing learning and improving predictive accuracy. This study examines spatially explicit *K*-nearest neighbors (SE-KNN) by integrating SA as a spatially explicit property, λ, into the learning algorithm. The innovation of SE-KNN lies in its alignment with the principle of spatial autocorrelation, as KNN's core learning assumption—that similar observations tend to have similar outcomes—naturally parallels spatial dependence. The proposed SE-KNN is applied to a house price prediction model using house sales data from Franklin County, Ohio to demonstrate a spatially dependent, data-rich, and real-world problem. The results show that SE-KNN achieved the best prediction accuracy compared to mean of absolute error (MAE) of three other machine learning approaches (i.e., standard KNN, linear regression, and artificial neural networks). The proposed method effectively captures the spatial structures in the housing market and leaves only a trace amount of SA in the residuals.

Keywords: spatial machine learning; spatial autocorrelation; spatially explicit KNN; house price prediction; attribute distance; spatial distance

Check for updates

Academic Editors: Wolfgang Kainz, Dev Raj Paudyal, Jingzhong Li and Binbin Lu

Received: 25 November 2025
Revised: 13 January 2026
Accepted: 17 January 2026
Published: 21 January 2026

Copyright: © 2026 by the authors. Published by MDPI on behalf of the International Society for Photogrammetry and Remote Sensing. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Machine learning (ML) furnishes a flexible approach in solving problems in various research fields and domains. It has gained popularity in geography or geospatial information sciences (GIS) research as well. For instance, recent research suggests ML models are capable of handling sizable georeferenced data across diverse fields, such as geography, the environment, and urban planning [1]. However, some ML models using spatial data show a lack of consideration of spatial autocorrelation [2]. Nikparvar and Thill [3] emphasize that spatial data represent a non-trivial class of data characterized by spatial dependence, spatial heterogeneity, and scale effects. When these fundamental properties are not explicitly addressed in ML models, both model learning and predictive performance may be adversely affected.

In response to this limitation, spatial ML (spatially explicit ML), often referred to as GeoAI, has emerged as an active area of research that incorporates intrinsic spatial characteristics into machine learning algorithms [1,4–6]. For example, Hengl et al. [7]

proposed a spatially enhanced random forest model by incorporating buffer-based distance measures as geographic covariates, while Das et al. [8] introduced a housing price prediction framework based on graph neural networks that embed the spatial context of neighborhood amenities. In contrast, the potential of K-nearest neighbors (KNN) models augmented with spatial features has received relatively limited attention in comparison to the more complex spatial ML models above although KNN is a widely used and effective algorithm in many domains.

The KNN method learns that observations close in attribute distance are likely to have similar output values and generalizes to unseen data. Specifically, KNN identifies the K nearest neighbors of a data point based on distance in an attribute space, assuming that nearby points in this multivariate feature space are likely to have similar outcomes or labels. This assumption is like spatial autocorrelation (SA). Positive SA (PSA) describes a spatial pattern in which observations close in geographic space tend to be similar on a two-dimensional surface [9]. That is, the fundamental idea of KNN has an analogy to the concept of SA.

Despite its algorithmic simplicity, KNN has demonstrated good performance across a wide range of applications and is widely regarded as an effective method for missing data imputation in machine learning [10]. Given the prevalence of incomplete observations in data-driven analyses, approaches for handling missing values are essential for achieving reliable analytical outcomes. Another advantage of KNN is flexibility. It can be applied to all data distributions, not limited to common statistical distributions such as the Gaussian or Poisson [11].

Therefore, while retaining their advantages, such as effective imputation performance and flexibility in an attribute space and acknowledging that most socio-economic data exhibit a significant level of PSA [12,13], SA incorporation in KNN can furnish a potential for gaining meaningful geospatial insights into the learning process [1].

This study introduces a modified KNN framework that integrates both attribute-based and spatial features into the similarity assessment in order to enhance predictive accuracy. The proposed method, referred to as spatially explicit K-nearest neighbors (SE-KNN), evaluates proximity using two complementary distance components: a non-spatial attribute distance, consistent with standard KNN, and a spatial distance reflecting geographic proximity. These components are combined through a mixing parameter λ, which controls their relative influence when identifying the K nearest neighbors, ensuring that selected neighbors are similar in both housing characteristics and geographic location. As a result, SE-KNN is designed to jointly capture attribute relationships and spatial structure within the learning process.

2. Background

2.1. Spatially Explicit Machine Learning Models

As mentioned above, while spatial dependence in ML models based on spatial data has received greater attention, the development of methods has diverged into two broad approaches [2]. The first approach focuses on the direct inclusion of spatial features within ML models. For example, Hengl et al. [7] introduced geographical covariates such as buffer distances into an RF model to mimic spatial dependencies typically captured by geostatistical methods. Li et al. [14] developed a geointelligent deep belief network using spatially and temporally lagged PM2.5 variables implemented through stacked restricted Boltzmann machines (RBMs). Similarly, Kiely and Bastian [15] reported improved predictive accuracy in real estate modeling by integrating spatial lag features. Liu et al. [2] showed that spatial lags and eigenvector spatial filtering (ESF) could enhance RF predictions for housing prices, with LASSO feature selection helping to balance model complexity. Das

et al. [8] introduced a graph-neural-network-based housing price prediction framework that learns geospatial network embeddings (GSNEs) for houses and points of interest in a weighted multipartite graph, capturing both local neighborhood relationships (e.g., houses and nearby schools or transit stations) and global similarities in amenity context across locations.

Second, there are hybrid approaches that combine ML models with other functions, such as geostatistical methods. Behrens et al. [16] enhanced spatial prediction for digital soil mapping by incorporating Euclidean distance fields (EDFs) in ML as spatial covariates, allowing ML models such as a linear model, bagged multivariate adaptive regression splines (MARSs), radial basis function support vector machines (SVMs), Cubist, RF, and a neural network (NN) ensemble to internalize spatial structure without separate residual correction. Chen et al. [17] applied geographically weighted regression kriging and artificial neural network kriging to predict soil organic carbon, showing that integrating SA directly into the modeling process led to reduced error and improved performance. Foresti et al. [18] proposed combining kriging with a neural network for precipitation estimation. Geoganos et al. [19] implemented geographical random forests (GRFs) by explicitly accounting for spatial heterogeneity by fitting multiple localized RF models based on spatial proximity via a kernel or bandwidth. Local predictions are then combined with a global RF via a weighting parameter to balance local and global effects.

Collectively, as summarized in Table 1 for the studies above, most existing methods either treat SA in a discrete manner through feature engineering or localized model structures or impose it continuously via distance-based or geostatistical formulations. There remains room to consider balancing between spatial dependence and attribute-based explanatory variables. In this context, the KNN approach represents an alternative for developing spatially explicit machine learning models.

Table 1. Approaches for incorporating spatial dependence in machine learning models.

Approach	Study	Machine Learning Models	Spatial Mechanism	Treatment of Spatial Autocorrelation (SA)
Direct inclusion of spatial features	Liu et al. [2]	Random forest (RF)	Spatial lags and Moran eigenvectors	Spatial lag
	Li et al. [5]	Stacked restricted Boltzmann machines (RBMs)	Spatially and temporally lagged variables	Spatial lag
	Kiely and Bastian [15]	RF, Generalized linear model (GLM), Gradient boosting model(GBM), Artificial neural network (ANN)	Spatially lagged features	Spatial lag
	Hengl et al. [7]	RF	Geographical covariates (buffer-distance variables)	Discrete [1]
	Das et al. [8]	Graph neural network (GNN)	Geospatial network embeddings capturing neighborhood relationships	Discrete
Hybrid approach	Behrens et al. [16]	MARS, SVM, Cubist, RF, NN	Euclidean distance fields (EDFs) as spatial covariates	Continuous [2]
	Chen et al. [17]	ANN	Geographically weighted regression kriging	Continuous
	Foresti et al. [18]	NN	Kriging–NN hybrid modeling	Continuous
	Geoganos et al. [19]	Geographical Random forest (GRF)	Localized RF models using spatial kernels/bandwidths	Discrete

[1,2] Discrete indicates that SA is represented using binary or categorical structures (e.g., spatial adjacency, neighborhood membership, or localized model partitions), whereas continuous denotes approaches that model the magnitude of SA as a smoothly varying function of distance or spatial proximity from each observation.

2.2. Spatially Explicit Approach to KNN

KNN has been widely used in pattern recognition and statistical estimation since the 1970s [20]. Despite its conceptual simplicity, KNN has demonstrated strong performance across a range of geospatial applications. With respect to incorporating geographical context into KNN models, existing studies can be broadly classified into three categories: (1) standard KNN models with locational information, (2) distance-weighted KNN, and (3) spatially constrained KNN.

The first category, standard KNN with locational information, represents a spatially explicit machine learning approach based on the direct inclusion of spatial features. This strategy is the most commonly adopted in spatially explicit KNN applications. For example, Baldominos et al. [21] developed a ML framework for housing price prediction and investment opportunity identification. It shows that KNN outperformed several alternative models, including support vector regression and multilayer perceptrons. Hagenauer et al. [22] evaluated 38 ML models to predict municipal-level land consumption in Bavaria, Germany, for the period 2015–2021. Although KNN was not the top-performing model, it achieved competitive results and surpassed both single-tree methods (e.g., ctree and rpart) and linear models (e.g., lasso and glm). In the context of missing data imputation, Sanjar et al. [23] applied KNN to geolocation-based price prediction, reporting gains in both computational efficiency and predictive accuracy.

Despite these successes, such approaches typically treat spatial coordinates as ordinary attributes and do not explicitly model spatial dependence, which may limit KNN performance by relying solely on attribute-based similarity within the distance function.

The second category, distance-weighted KNN, extends the standard KNN framework by weighting neighbors according to their relative proximity within a predefined bandwidth. Dudani [24] first proposed a distance-weighting scheme in which the nearest neighbor receives full weight, the farthest neighbor receives zero weight, and intermediate neighbors are weighted smoothly between these extremes. Atkinson [25] applied distance-weighted KNN in land-cover classification by allowing pixel labels to depend jointly on spectral similarity and geographic proximity. This approach yielded modest but consistent improvements in classification accuracy on simulated imagery. While distance-weighted KNN introduces a continuous notion of spatial proximity, it does not explicitly balance the influence of spatial dependence relative to attribute-based similarity.

The third category, spatially constrained KNN, has primarily been developed in computer vision and image retrieval contexts, where target objects may appear under varying viewpoints, scales, and cluttered backgrounds. Shen et al. [26] proposed a spatially constrained similarity measure that rewards only geometrically consistent local feature matches in terms of position, scale, and orientation. This spatially constrained similarity score is then used within a KNN framework for re-ranking and object localization in large-scale image databases. Although this approach explicitly enforces spatial consistency, it is designed for geometric matching problems and is less directly applicable to spatial dependence in geographic prediction tasks.

In summary, the predominant strategy for incorporating locational information into KNN has been the direct inclusion of spatial features as additional variables. This practice originates from the mainstream computer science literature [27]. The main drawback of the first category is that, when the input dataset has many variables, the influence of spatial information diminishes because equal weights are given to all input variables. Hence, as more variables are included, the prominence of spatial information reduces. This procedure lacks the flexibility to control weights between spatial and non-spatial features. A flexible approach to balancing the inherent spatial pattern and attribute-based similarity can further enhance the capability of KNN to incorporate spatial dependence.

3. Methodologies

KNN is known for its flexibility, intuitiveness, and easy tuning [28]. It is a flexible supervised learning approach that performs effectively on large datasets characterized by complex underlying data-generating mechanisms. In addition, the method is conceptually straightforward and easy to interpret. Moreover, unlike many popular ML methods such as SVM or RF, KNN involves a small number of hyperparameters, primarily the number of nearest neighbors, K, which directly controls the bias–variance tradeoff and can be tuned straightforwardly [29]. This study introduces the standard KNN method and the SE-KNN by using house sales data from Franklin County, Ohio (detailed information of the dataset is discussed in Section 4.1 later). On one hand, the KNN method can be described as follows [30]:

For n observations and p variables in input data:

(1) Calculate the distance d_{ij} between observation i (i = 1, 2, …, n) and observation j (j = 1, 2, …, n);
(2) Sort the distances d_{ij} in ascending order, and acquire the first K indices;
(3) Get K neighbors' response values and return the mean of the values as the prediction.

The KNN method involves three main steps. In step (1), it calculates the Euclidean distance d_{ij} using the p-dimensional predictors (equal to the number of variables), measuring the attribute dissimilarity between observations:

$$d_{ij} = \left\| x_i - x_j \right\|_2 \quad (1)$$

where observations x_i and observation x_j are p-dimensional vectors of predictor variables for observations i and j, respectively; and $\|\cdot\|_2$ denotes the Euclidean norm.

In step (2), it finds the K nearest neighbors by ranking the distance values. The K is a hyperparameter tuned by finding the lowest test error in a validation set. Choosing an optimal K value is crucial in KNN since it determines the bias–variance tradeoff of the method. As K increases, the bias of the estimates increases, but the variance decreases. In step (3), it estimates values by averaging neighbors' values in the response variable as a mean $y_i = \sum_{t=1}^{K} y_t / K$ where y_i denotes a predicted response variable for observation i, K is the chosen number of nearest neighbors, y_t is the known response variable of the observation t in the K nearest neighbor set.

On the other hand, SE-KNN is an extension of the standard KNN method. In order to take advantage of spatial patterns or SA among neighboring locations, spatial distances are incorporated into the standard KNN framework. Recent studies have shown that adapting similarity measures in ML models to reflect spatial structure can substantially improve predictive performance in spatial modeling contexts [31], and broader reviews in geospatial ML further emphasize the necessity of explicitly accounting for spatial dependence to avoid biased inference and misleading model evaluation [32]. Building on this insight, this study introduces λ as an additional hyperparameter to balance the effect of spatial autocorrelation with attribute similarity. The detailed procedure of SE-KNN can be expressed as follows:

For n observations, p attribute variables, and n sets of q-dimensional coordinates as spatial variables in the input spatial dataset:

(1) Separate inputs into p attribute variables and q spatial variables, and calculate two distances: spatial distance and attribute distance between observation i (i = 1, 2, …, n) and observation j (j = 1, 2, …, n);
(2) Combine the two distances into one distance metric d'_{ij} using a tunable hyperparameter λ;
(3) Sort the distances d'_{ij} in ascending order, and acquire the first K indices;
(4) Get K neighbors' response values and return the mean of the values as the prediction.

The distance formulation used in SE-KNN differs from that of the standard KNN defined in Equation (1). To identify the K nearest neighbors, SE-KNN modifies the distance computation to jointly account for attribute-based dissimilarity and spatial proximity. Specifically, spatial and non-spatial features are treated separately, two corresponding distance measures are computed, and these measures are combined through a mixing parameter $\lambda \in [0, 1]$ to obtain the composite distance d'_{ij} used for neighbor selection:

$$d'_{ij} = (1-\lambda)\left\|x_i - x_j\right\|_2 + \lambda\left\|\mu_i - \mu_j\right\|_2 \quad (2)$$

where μ_i and μ_j denote the q-dimensional spatial coordinate vectors (e.g., projected x–y coordinates) for observations i and j; λ is a tunable hyperparameter to set a ratio between attribute and spatial distances.

SE-KNN introduces a tunable hyperparameter, λ, to explicitly regulate the relative influence of spatial proximity and attribute-based similarity within the learning process, rather than treating geographic coordinates as ordinary non-spatial predictors. At one extreme, when $\lambda = 0$, the method reduces to the standard KNN and relies solely on attribute information; at the other extreme, when $\lambda = 1$, neighbor selection is driven entirely by spatial distance, independent of attribute similarity.

The motivation for this formulation is that spatial data are managed by two distinct but interrelated processes. One process is non-spatial, reflected in relationships among attribute variables, such as the association between housing characteristics (e.g., number of bedrooms) and sale prices. The other process is spatial, manifested through geographic structures in the data, such as price similarities among nearby houses. Accordingly, SE-KNN is parametrized by two intuitive hyperparameters, λ and K, both of which can be efficiently tuned within a supervised learning framework using a validation dataset.

4. Empirical Application

This section presents the empirical application of the SE-KNN method and other ML methods for predicting house prices and explains the data preparation process and model configurations used in the analysis. This section first describes how the input data are preprocessed and standardized to ensure consistency across approaches. It then examines the SA present in the dataset to demonstrate the importance of incorporating spatially explicit components into predictive models. Lastly, the section outlines how the models are implemented in this study, including SE-KNN and other models, such as standard KNN, linear regression (LR), and artificial neural network (ANN).

4.1. Dataset

House prices are intrinsically spatial since they are affected by geographic locations as well as house characteristics. That is, the commodity of a house is a bundle of internal characteristics and external locational factors. Studies (e.g., [33–35]) discuss that such characteristics can be categorized with the following three groups: physical features, locational characteristics, and neighborhood and available amenities. Houses located in the same neighborhood or in proximity often exhibit similar styles, structural characteristics, and construction years, and they generally access similar public services. In practice, real estate valuation commonly relies on identifying recently sold properties with comparable characteristics located within close proximity. As a result, housing prices are shaped not only by intrinsic property attributes but also by the prices of nearby houses, reflecting a spatial spillover effect. This valuation practice closely corresponds to the underlying rationale of the SE-KNN approach.

This study analyzes the empirical dataset of one-year Franklin County house sales (from 1 January 2019 to 31 December 2019), which was obtained from the Franklin County

Auditor as open records. Franklin County, Ohio is the center of the Columbus Metropolitan Statistical Area (MSA), situated in the capital of Ohio. According to the US Census Bureau, Franklin County is the most populous county in Ohio, and the 2015–2019 median value of owner-occupied houses is USD 175,100. During the study period, 18,843 residential property transactions without repeat sales were initially identified. After data preprocessing, 16,588 house sales were retained for subsequent analysis following the removal of abnormal records and outliers. Excluded observations consisted of land-only transactions, parcels exceeding 5 acres, properties with zero bedrooms, homes with total living areas smaller than 200 square feet or larger than 1000 square feet, and sales prices below USD 52,500, which corresponds to the 5th percentile of the overall price distribution. This lower-tail trimming was implemented to eliminate extreme low-price transactions, as housing prices are highly heterogeneous and the lower tail often exhibits behavior distinct from the central portion of the distribution [36]. The response variable used for prediction is the house sale price, expressed in US dollars.

To construct the input dataset, several preprocessing steps are applied, including feature transformation, feature selection, imputation, and scaling. First, during feature transformation, categorical variables are converted into numeric form so they can be used in models that cannot directly incorporate text data. Because the dataset contains categorical variables with a natural rank order, ordinal encoding is used. For example, the house condition variable is encoded as integers from 1 to 8, corresponding to the categories excellent, very good, good, average, fair, poor, unsound, and very poor. Second, in the feature selection stage, predictors with zero or near-zero variance, or those with more than 50% missing values, are removed. Additional feature selection is performed by retaining predictors with absolute correlation coefficients of at least 0.1 with the response variable. Third, variables with missing values are imputed using median values and then standardized using their means and standard deviations, since both KNN and ANN are sensitive to differences in feature scales. Finally, following the optimal data-splitting ratio recommended by Joseph [37], the full dataset of 16,588 house sales is randomly divided into training (64%, 10,616 observations), validation (16%, 2654 observations), and testing (20%, 3318 observations) sets.

To summarize, the SE-KNN model uses fifteen house-characteristic variables for predicting *sales prices*, consisting of thirteen non-spatial predictors (twelve numerical variables and one categorical variable) and two spatial predictors. The thirteen predictors are *the number of total rooms, bedrooms, dining rooms, full baths, half baths, stories, total fixtures, additional fixtures, total living area, total land area, house age, sales month,* and *house condition.* The two spatial predictors are the projected *x* and *y coordinates* of each house, where its projection is Ohio South State Plane Coordinates System (FIPS: 3402 and unit: US foot). Figure 1 presents the selected predictors and the response variable along with their pairwise correlation coefficients. The results show that the *x coordinate* and *house age* have negative correlations with *sales prices*, whereas all other predictors display positive correlations.

4.2. Spatial Autocorrelation in the House Sales Prices Data

Figure 2 illustrates the spatial distribution of house sale transactions in the training and validation datasets. House prices are encoded using a color gradient ranging from extremely high values (dark red) to extremely low values (dark green). At the global scale, higher house prices and greater transaction densities are concentrated in the northern portion of the study area, while lower prices and fewer transactions are observed in the southern region. At a finer spatial scale, distinct clusters of high–high and low–low house values appear across the map, suggesting the presence of strong positive spatial autocorrelation in housing prices.

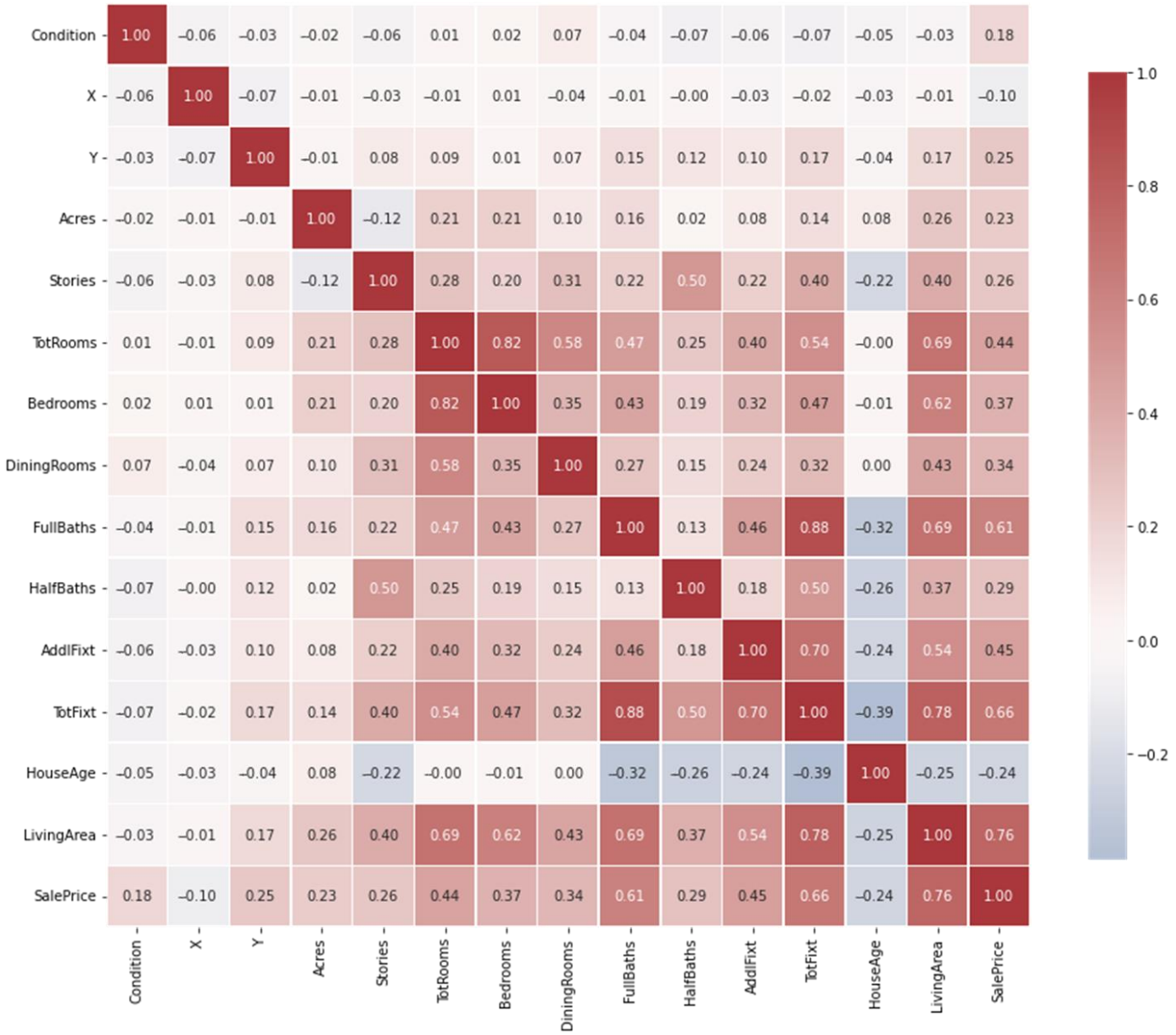


Figure 1. Pairwise correlation coefficients matrix of the dataset.

This spatial dependence is further confirmed through both visual and statistical diagnostics. Figure 3 presents the Moran scatterplot, which plots house prices against the row-standardized spatial lag of neighboring prices, providing a graphical depiction of the strength and direction of spatial autocorrelation. The positive slope of the fitted regression line indicates strong PSA. Consistently, the global Moran's I value, computed using a distance-based eight-nearest-neighbor spatial weights matrix, is 0.6904 with a p-value of 0.000, indicating statistically significant SA.

In practical terms, these results reflect pronounced geographic clustering of similar house values. While SA does not inherently violate the assumptions of many ML algorithms, failing to account for such spatial structure can lead to biased or overly optimistic model evaluations [1,38–41]. An effective predictive model should capture systematic spatial patterns present in the data and leave only irreducible random noise in the residuals. If residuals exhibit spatial clustering, such as consistently larger errors in specific areas, this suggests unmodeled spatial structure. Consequently, Moran's I statistics computed on

model residuals provide a useful diagnostic tool for assessing whether a machine learning model adequately addresses spatial dependence.

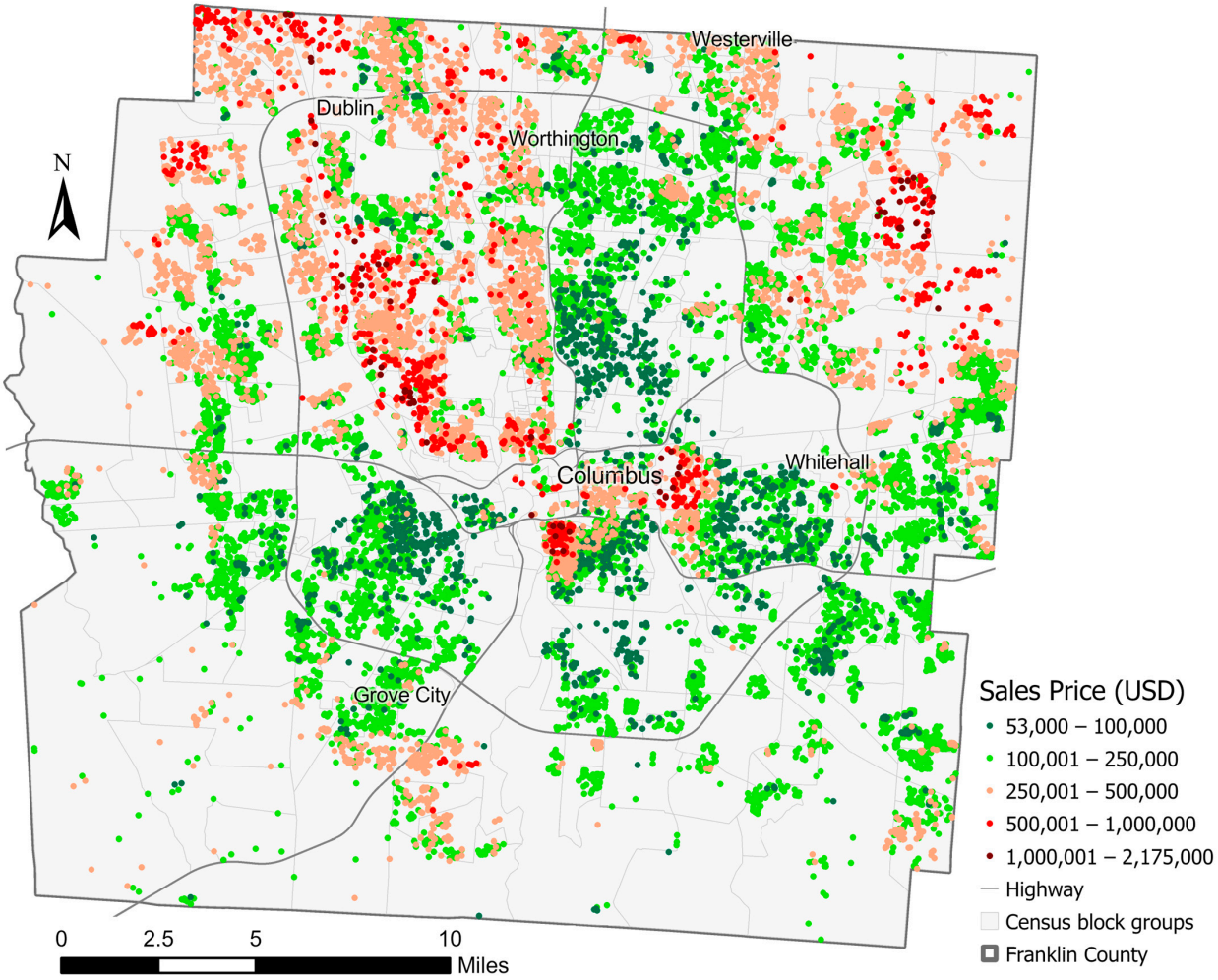


Figure 2. Map of house prices in Franklin County, OH, from 1 January 2019 to 31 December 2019.

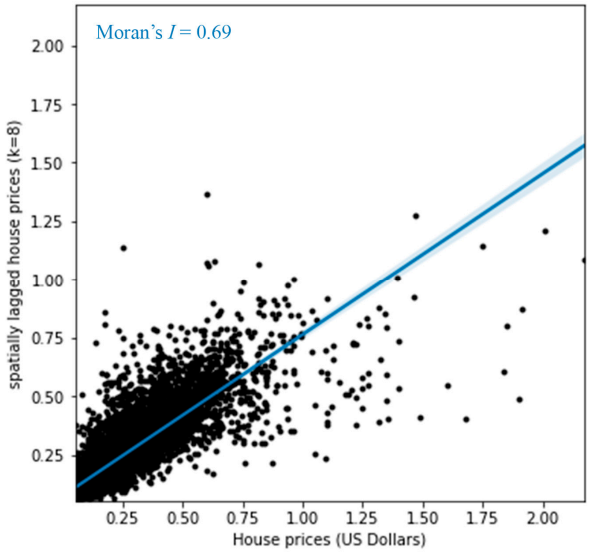


Figure 3. Moran scatterplot of house prices (n = 13,270). Note that the slope of the blue regression line represents the Moran's I statistic (I = 0.69), and the shaded blue region indicates its 95% confidence interval.

4.3. House Sales Prices Prediction Models and Performance Evaluation

To evaluate the effectiveness of SE-KNN for spatial prediction, its performance is compared against three non-spatial machine learning models: standard KNN, LR, and ANNs. Model performance is assessed using four criteria: root mean squared error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and residual SA measured by Moran's I.

Standard KNN serves as the baseline model against which the proposed extension is evaluated. LR is included as a widely used benchmark model that provides a reference for assessing linear relationships between the response variable and predictors. ANN represents a flexible non-linear learning approach that has gained increasing attention in recent machine learning studies and has been applied to housing price prediction in prior work [42,43]. The standard KNN, LR, and ANN models are implemented using the Scikit-learn library, while the SE-KNN algorithm is implemented in Python 3.12.

For model evaluation, RMSE and MAE are used to quantify prediction error magnitudes. MAE reflects the average absolute error by assigning equal weight to all deviations, whereas RMSE penalizes larger errors more heavily. Both metrics range from zero to infinity, with lower values indicating better predictive performance. In addition, R^2 is computed to measure the proportion of variance in observed values explained by each model. To assess whether spatial dependence remains unmodeled, Moran's I is calculated on the residuals of each model, with values close to the expected value indicating minimal remaining SA. RMSE, MAE, and R^2 are computed using built-in Scikit-learn functions, while Moran's I is calculated using the PySAL spatial analysis library.

5. Results

This section presents the results of tuning and evaluating the SE-KNN model using the 2019 Franklin County house sales data. First, the optimal values of the two hyperparameters of SE-KNN—the spatial ratio λ and the number of neighbors K—are identified through a grid search based on the validation dataset. During this process, 1000 replicated simulation runs are examined to determine which of the two hyperparameters has a stronger influence on model performance and, therefore, should be prioritized. Next, the SE-KNN model with its optimal hyperparameters is compared against three non-spatial machine learning models—KNN, LR, and ANN—using the testing dataset. Finally, the spatial patterns of predictions and residuals are analyzed to assess whether the model outputs align with the known urban structure of Franklin County, OH and to evaluate how effectively SE-KNN accounts for spatial autocorrelation.

5.1. SE-KNN Hyperparameters: The Spatial Ratio λ and the Number of Neighbors K

SE-KNN uses two parameters: the spatial ratio λ and the number of nearest neighbors K. To tune the SE-KNN model, a grid search was performed over all combinations of λ (0 to 1 in increments of 0.1) and K (2 to 19 in increments of 1). This procedure identifies the hyperparameter set that yields the highest predictive accuracy when evaluated on the validation dataset. The grid search was repeated 1000 times to assess the variability in model performance. The results, summarized in Figure 4, show a consistent decline in MAE as λ increases from 0 to approximately 0.7. This indicates that incorporating spatial information improves predictive accuracy up to this threshold. However, when λ is set to 0 (using only attribute distance) or 1 (using only spatial distance), the MAE values are substantially higher compared to settings in which both attribute and spatial distances are jointly integrated.

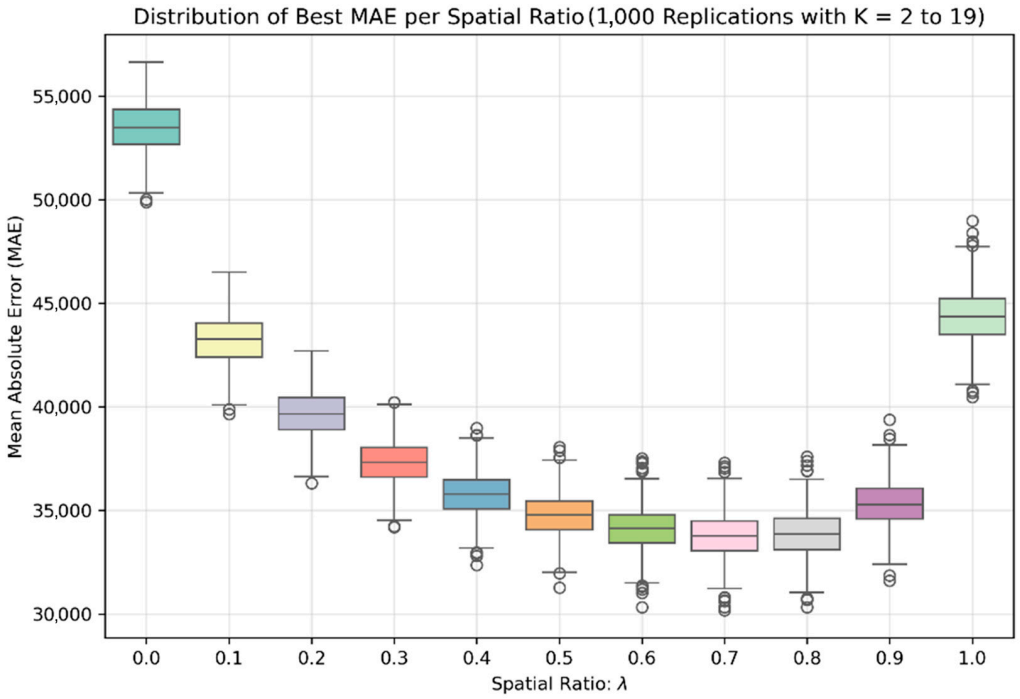


Figure 4. Distribution of the lowest MAE values from 1000 SE-KNN simulation replications (K = 2 to 19), representing the best-performing K for each spatial ratio λ. Note that circles are beyond 1.5 interquartile range.

Table 2 presents the best K values and the descriptive statistics of the simulation results that produce the lowest error metrics for each spatial ratio λ from 0 to 1. Based on the mean RMSE and MAE values, $\lambda = 0.7$ and $K = 6$ yield the best performance overall, with the lowest errors (RMSE = 64,564; MAE = 33,752) and the highest R^2 (0.825). Accordingly, this parameter set is adopted as the final SE-KNN model configuration for comparison with the other machine learning methods.

Table 2. The best K and model performance of SE-KNN depending on spatial ratio λ with 1000 replications. (unit: US dollar for RMSE and MAE).

λ	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Best K [1]: median	14	10	10	9	9	8	7	**6**	5	5	7
Best K: min–max	6–19	4–19	4–19	4–19	4–19	4–17	4–13	3–11	3–9	3–8	3–12
RMSE: mean	86,723	75,685	71,029	68,257	66,456	65,188	64,703	**64,564**	65,188	67,990	84,145
RMSE: S.D.	4129	4338	4435	4558	4687	4778	4906	4983	5021	5076	5468
MAE: mean	53,497	43,241	39,660	37,309	35,774	34,739	34,113	**33,752**	33,851	35,308	44,351
MAE: S.D.	1211	1134	1080	1041	1034	1019	1030	1037	1053	1092	1277
R^2: mean	0.684	0.759	0.788	0.804	0.814	0.821	0.824	**0.825**	0.821	0.805	0.702
R^2: S.D.	0.018	0.018	0.017	0.017	0.017	0.018	0.018	0.019	0.019	0.020	0.025

[1] Best K refers to the value of K that produced the lowest MAE for each spatial ratio. Note that figures in bold indicate the best-performing metrics.

Furthermore, SE-KNN incorporates two hyperparameters, whereas standard KNN relies on only one K. When selecting hyperparameters, situations may arise in which improving one parameter worsens the performance associated with the other. In such cases, it becomes important to understand which parameter should be prioritized. To address this issue, simulation results were analyzed to determine which parameter exerts

a stronger influence on model performance. Figure 5 illustrates the variability in model accuracy when one parameter is held constant. On one hand, in Figure 5a, when λ is fixed at 0.7, MAE ranges from 30,175 to 33,309, and the two lowest MAE values (30,175 and 30,260) occur when K equals 6 or 8. On the other hand, in Figure 5b, when K is fixed at 6, MAE varies more widely—from 30,175 to 53,083. And its lowest value occurs at $\lambda = 0.7$. These results indicate that model performance is far more sensitive to changes in λ than to changes in K. Therefore, when tuning SE-KNN, identifying an appropriate λ value should be given a higher priority.

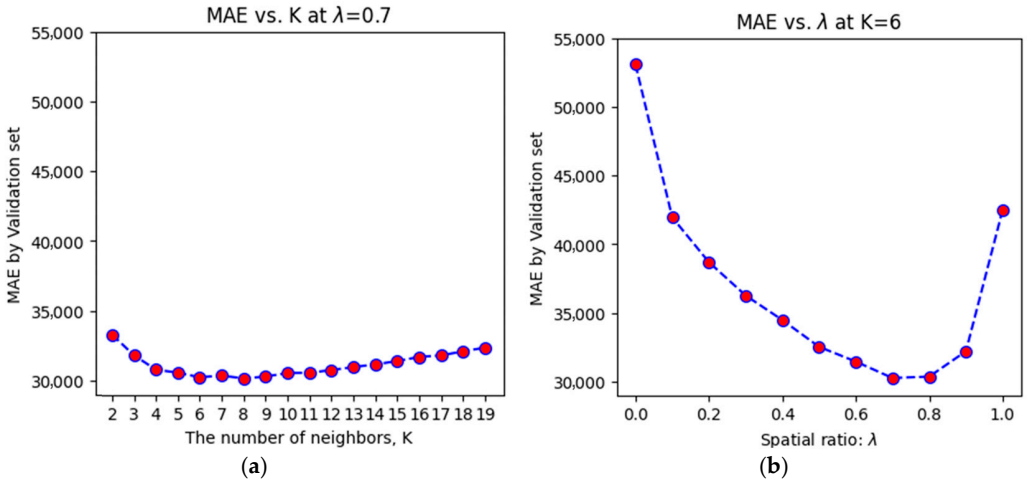


Figure 5. Comparison of the effects of the two hyperparameters on model performance. (**a**) MAE across K values when the spatial ratio is fixed at $\lambda = 0.7$. (**b**) MAE across spatial ratio λ values when the number of nearest neighbors is fixed at $K = 6$.

5.2. Comparison of Prediction Accuracy

Table 3 summarizes the prediction performance of SE-KNN relative to three non-spatial machine learning models: standard KNN, LR, and ANNs, using RMSE, MAE, R^2, and Moran's I of residuals. To ensure a fair comparison, all models were tuned using the same validation dataset to determine their optimal hyperparameters. Overall, SE-KNN demonstrates the strongest performance, followed by ANN, standard KNN, and LR.

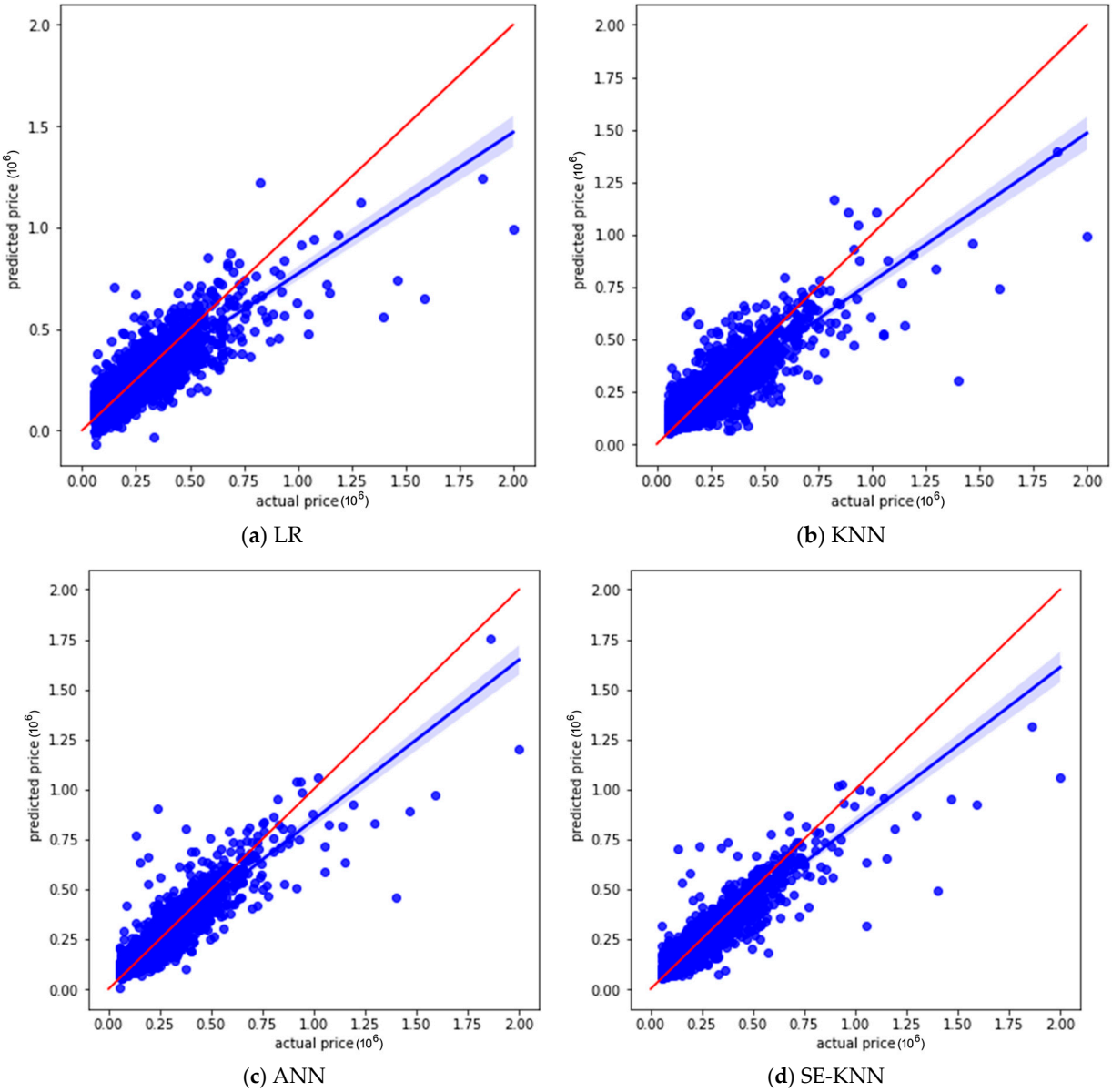
Table 3. Comparison of Prediction metrics for LR, KNN, ANN, and SE-KNN Methods.

Methods	RMSE	MAE	R^2	Moran's I	Hyperparameters
LR	85,542	56,782	0.682	0.366	None
KNN	77,366	43,984	0.739	0.255	K = 10
ANN	64,981	37,969	0.816	0.113	Hidden layers: (250, 120, 60, 10); activation: ReLU
SE-KNN	64,609	32,571	0.822	0.025	$\lambda = 0.7$; $K = 6$

Among the four models, LR exhibits the weakest predictive capability, yielding the highest error values (RMSE = 85,542; MAE = 56,782) and the lowest explanatory power ($R^2 = 0.0682$). In addition, the residuals from the LR model retain a moderate level of spatial autocorrelation (Moran's $I = 0.3668$), indicating that substantial spatial structure remains unaccounted for. Standard KNN (with $K = 10$) improves upon LR by reducing prediction errors (RMSE = 77,366; MAE = 43,984), increasing explanatory power ($R^2 = 0.739$), and lowering residual spatial autocorrelation (Moran's $I = 0.255$). These results suggest that similarity-based learning captures non-linear relationships more effectively than a purely linear specification.

The ANN model further improves predictive accuracy relative to LR and standard KNN. When configured with four hidden layers (250, 120, 60, and 10 neurons) and ReLU activation functions, ANN achieves lower error values and higher R^2, while mitigating overfitting through network depth. Nevertheless, the proposed SE-KNN model, with optimal hyperparameters ($\lambda = 0.7$, $K = 6$), outperforms all competing approaches. SE-KNN achieves the lowest RMSE (64,609) and MAE (32,571) and the highest explanatory power ($R^2 = 0.822$) and leaves only a negligible amount of spatial autocorrelation in the residuals (Moran's $I = 0.025$).

Additional diagnostic evidence is provided through scatterplots in Figures 6 and 7. Figure 6 presents observed versus predicted value plots for LR, KNN, ANN, and SE-KNN, with the 1:1 reference line shown in red. Points closer to this line indicate more accurate predictions. Among the four models, SE-KNN exhibits the tightest alignment with the reference line, indicating superior predictive accuracy.



(**a**) LR (**b**) KNN

(**c**) ANN (**d**) SE-KNN

Figure 6. The scatterplots of actual versus predicted values. (**a**) linear regression, (**b**) k-nearest neighbors, (**c**) artificial neural network, and (**d**) spatially explicit k-nearest neighbors.

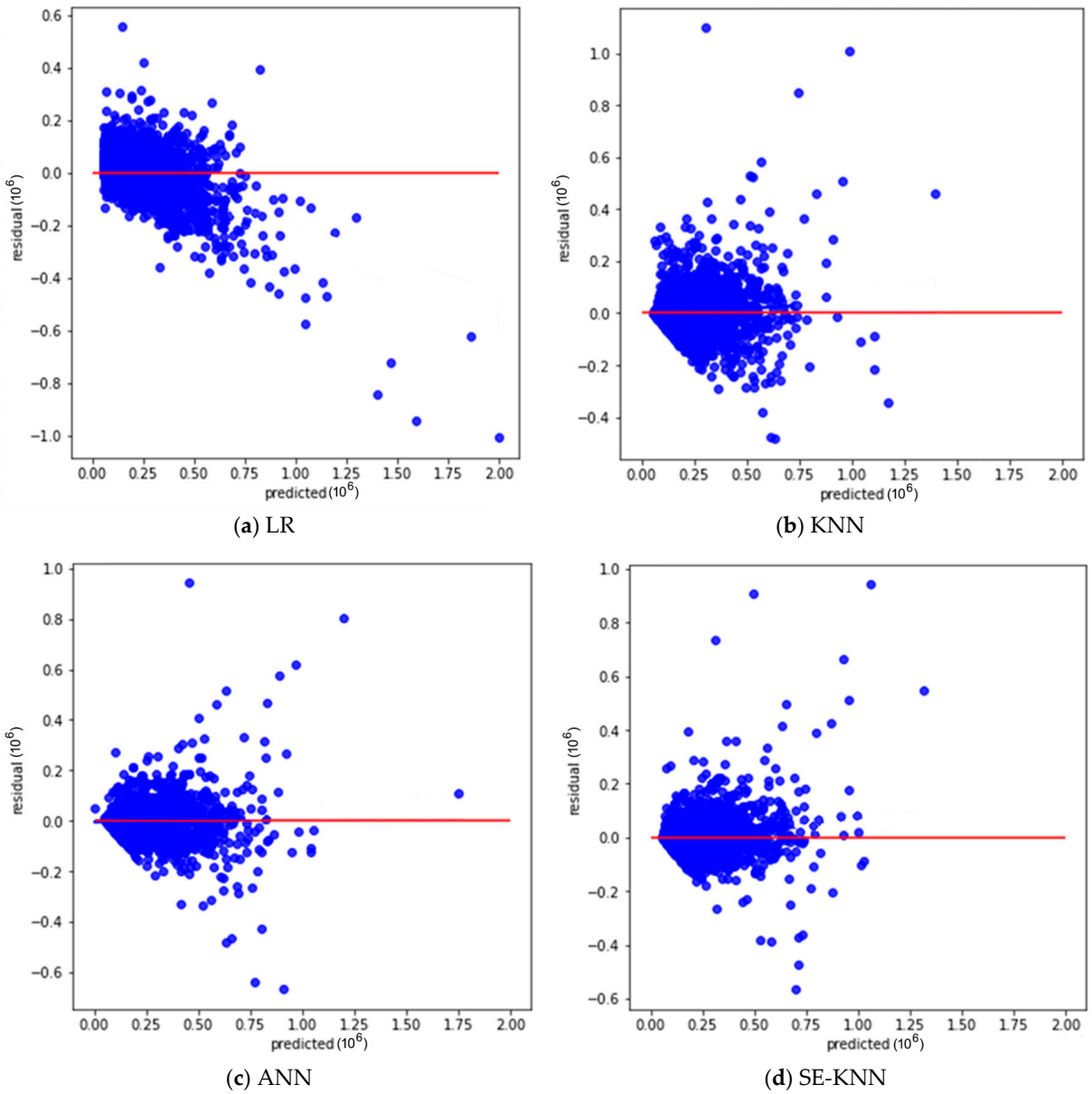


Figure 7. The scatterplots of predicted values against residuals with a reference line $y = 0$. (**a**) linear regression, (**b**) k-nearest neighbors, (**c**) artificial neural network, and (**d**) spatially explicit k-nearest neighbors.

Figure 7 displays scatterplots of predicted values versus residuals for each model, along with a horizontal reference line at zero. In Figure 7a, the LR model shows a pronounced linear pattern between predicted values and residuals. It suggests systematic structure remains unmodeled and indicates underfitting. The KNN model (Figure 7b) removes much of this linear pattern. However, residuals still exhibit a funnel-shaped distribution and notable outliers, despite reduced variance compared to LR. In contrast, the ANN and SE-KNN models (Figure 7c,d) produce residuals that are more symmetrically and compactly distributed around zero. This reflects a more random error structure and improved model adequacy.

5.3. Spatial Patterns of SE-KNN Results

The performance of the SE-KNN, which is the best model, can be further described with comparisons of the observed sales price map (Figure 2) and the predicted sales

price map generated by SE-KNN (Figure 8a). These maps show results for the test dataset (n = 3318). Based on the road network formed by the loop of Interstate 270 and the east–west corridors of Interstate 70 and 71 that extend north from central Franklin County, both maps exhibit similar spatial patterns. Specifically, areas between the northbound section of Interstate 70 and the westbound section of Interstate 71, including their intersections, show relatively high sales prices. In contrast, regions located within the inner loop of Interstate 270 and to the south of Interstate 70 tend to exhibit lower sales prices.

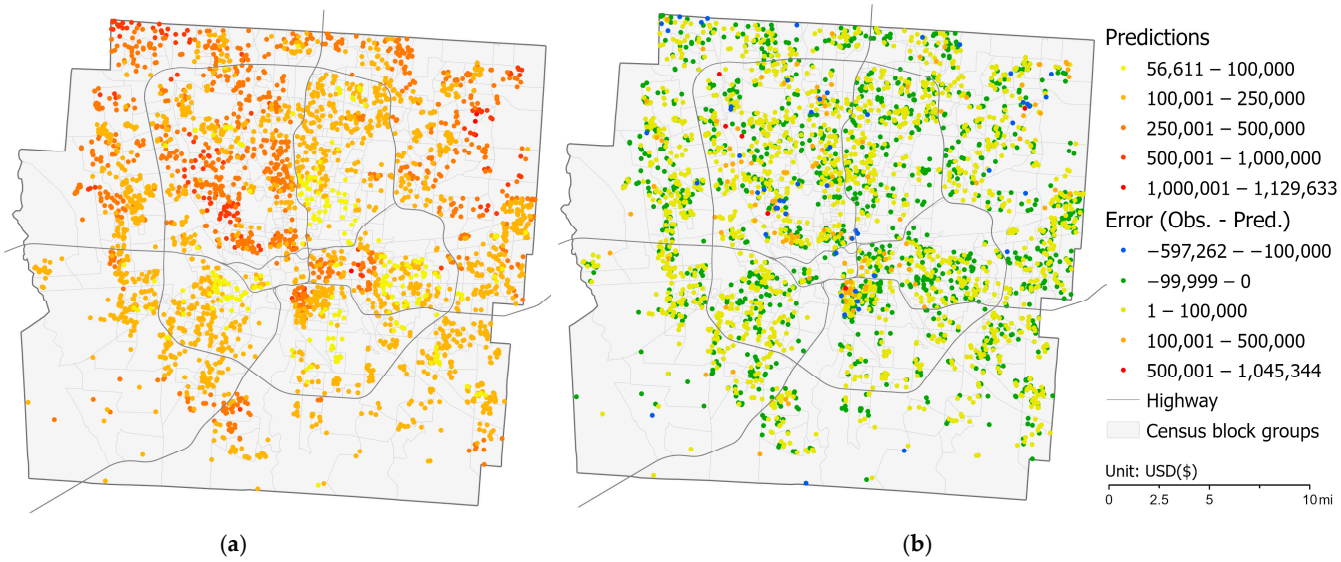


Figure 8. Spatial distribution maps of predictions (**a**) and errors (**b**) (n = 3318).

Regarding the spatial distribution of prediction errors (Figure 8b), the Moran's I statistic of 0.0741 indicates that the errors are approximately randomly distributed. This suggests that SE-KNN effectively accounts for spatial autocorrelation in the data. As aforementioned, this is a strength of SE-KNN, as ML models neglecting spatial dependencies often yield overly optimistic or biased predictions [38–41,44].

Since SE-KNN does not have a measure for support variable importance, Pearson's correlation coefficients are employed to assess the relative influence of each predictor on predictions and their errors, which are the discrepancies between observations and predictions. As shown in Table 4, both predicted values and the errors exhibit similar patterns of correlation with the predictors. Among all predictor variables, *living area* has the strongest positive correlation with predicted sales prices. This indicates that house size is a key determinant of market value. Similarly, other size-related variables, *the number of full bathrooms* and *bedrooms*, also show positive correlations with price. Home improvements, measured by the number of fixtures and additional fixtures, are also positively related to sales prices.

Table 4. The correlation table between predictors and both predictions and errors.

	Living Area	Fixtures	Number of Full Bathrooms	Number of Rooms	Additional Fixtures	Number of Bedrooms	X Coord	Y Coord
Predictions	0.76	0.68	0.62	0.47	0.47	0.39	−0.13	0.31
Errors	0.3	0.21	0.2	0.15	0.09	0.11	−0.03 [1]	0.01 [2]

[1] p-value is approximately 0.0545. [2] p-value is approximately 0.4589. All others without footnote are significant at $p < 0.0000$.

For prediction errors, positive correlations are observed with predictors—particularly *living area*, *full bathrooms*, and *fixtures*. This aligns with the finding that the SE-KNN model tends to produce larger errors for houses with extremely high sale prices. In contrast, spatial variables (x and y coordinates) show only weak or negligible correlations with both predictions and errors, suggesting that the spatial autocorrelation is already well-captured by the SE-KNN framework.

6. Discussion

Prediction models were applied to housing prices in Franklin County, where the study area exhibits strong positive spatial autocorrelation (Moran's $I = 0.69$). Performance evaluation indicates that the SE-KNN model outperforms the competing approaches. These results suggest several important implications regarding the advantages of incorporating spatial structure into machine learning models.

First, the empirical findings indicate that, under identical input conditions, SE-KNN yields superior predictive performance relative to several commonly used non-spatial machine learning methods for housing price estimation. Nevertheless, this outcome should be interpreted cautiously. The results primarily demonstrate that SE-KNN is advantageous when the response variable exhibits strong positive spatial autocorrelation, under which conditions it outperforms models such as linear regression and artificial neural networks. According to the well-established "no free lunch" theorem in machine learning, no single method consistently outperforms others across all problem settings, as model effectiveness is inherently data-dependent.

Second, SE-KNN can be an effective tool for interpolation or imputation, accounting for spatial autocorrelation. This advantage stems from the model's characteristics: it inherits the core mechanism of the standard KNN, which can predict interpolated or imputed values by averaging neighboring values, while also explicitly accounting for spatial proximity based on x and y coordinates.

Third, SE-KNN incorporates two hyperparameters—the spatial ratio λ and the number of nearest neighbors K. Across all 1000 simulation replications, λ emerges as the dominant factor influencing model performance, while changes in K have a comparatively modest effect. Accordingly, when tuning SE-KNN, selecting an appropriate λ should be prioritized. In the empirical application of the house sales prices in Franklin County, OH, the optimal model balances approximately 70% weight on spatial information with 30% on attribute-based (non-spatial) information. The resulting λ value of 0.7 underscores the substantial contribution of spatial factors to predictive accuracy and considers the spatial autocorrelation when developing models for house price prediction.

Finally, SE-KNN offers the advantage of being more intuitive and easier to tune than more complex machine learning models. Although ANN models can also achieve strong predictive performance for housing prices, they typically involve numerous interdependent hyperparameters and intricate network architectures, including learning rates, regularization parameters, numbers of hidden layers and neurons, optimization algorithms, activation functions, and training epochs. As a result, identifying a globally optimal hyperparameter configuration for ANN models is often challenging and computationally demanding.

However, in automated hyperparameter search settings, SE-KNN still requires explicit tuning during model training, as optimal hyperparameter values may vary with the degree of spatial autocorrelation present in the input data. In practice, a grid search with training and validation datasets can be employed as a standard machine learning procedure to identify a robust combination of hyperparameters that yields satisfactory predictive performance.

7. Conclusions

This study introduces a framework that incorporates both spatial distance and attribute-based dissimilarity into the KNN learning process to explicitly address inherent spatial autocorrelation and enhance predictive accuracy. The proposed SE-KNN can be regarded as an extension of the standard KNN that imposes spatial constraints through a distance-based weighting mechanism. Empirical results demonstrate that SE-KNN provides more accurate housing price predictions than comparable non-spatial machine learning models.

Given improved predictive performance, spatial ratio λ offers two important insights for spatially explicit ML. First, it provides an alternative strategy for mitigating the curse of dimensionality, in which increasing feature dimensionality leads to sparse data and less meaningful distance measures [45]. In geographic data, spatial autocorrelation often introduces redundant spatial information [9], and the parameter λ allows SE-KNN to explicitly control the influence of spatial dependence by balancing attribute similarity and geographic proximity. This flexibility enables the model to emphasize meaningful spatial similarity even when attribute-space distances become less informative.

Second, with respect to hyperparameter tuning, the joint effects of λ and K reveal that model performance is more sensitive to λ than to the number of neighbors K. Once an appropriate value of λ is identified, prediction accuracy becomes relatively stable across different values of K. It indicates that SE-KNN is comparatively easy to tune in practice. Taken together, these findings suggest that SE-KNN provides a practical and interpretable alternative for spatially explicit ML while it offers both improved predictive performance and meaningful spatial insight for tasks such as spatial interpolation and spatial prediction.

Another promising application of SE-KNN lies in spatial missing data imputation. When considering that KNN is widely used for imputing missing values in ML applications [10], SE-KNN is expected to provide an improved alternative for georeferenced datasets by explicitly accounting for spatial dependence, such as census population or socio-economic data. In addition, SE-KNN can be naturally extended to classification problems involving categorical response variables. In this case, the prediction rule is modified to assign the most frequently occurring class among the K nearest neighbors, rather than averaging continuous response values.

The applicability of SE-KNN and future study can be summarized in the following points. This study still requires greater external validity through applications to additional empirical cases. Accordingly, future work will examine how the prediction accuracy changes with different levels of SA, spatial scales, coordinate system, and density although spatial features are re-scaled. Furthermore, to achieve more robust validation and reduce potential sampling bias, although the SE-KNN model is tested with 1000 replications, future analyses will evaluate the proposed models using alternative data-partitioning strategies, such as spatial cross-validation and block-based spatial splitting. Finally, to further strengthen external validity beyond the current set of linear, non-linear, and neural-network-based comparison models, future research will expand comparative analyses to include additional spatial models, such as spatial lag models, geographically weighted regression, and spatial RF.

Author Contributions: Conceptualization, Meifang Chen; methodology, Meifang Chen; software, Meifang Chen and Changho Lee; validation, Changho Lee and Yongwan Chun; formal analysis, Meifang Chen; investigation, Meifang Chen and Changho Lee; resources, Meifang Chen; data curation, Yongwan Chun; writing—original draft preparation, Meifang Chen; writing—review and editing, Changho Lee and Yongwan Chun; visualization, Meifang Chen and Changho Lee; supervision, Yongwan Chun. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding. This research was supported by the University of Texas at Dallas and is part of the first author's PhD project entitled "SPATIALLY EXPLICIT MACHINE LEARNING APPROACHES FOR HOUSE PRICE MODELS".

Data Availability Statement: The original data presented in the study are openly available in Franklin County Auditor at https://www.franklincountyauditor.com/home (lastly accessed on 16 January 2026).

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Janowicz, K.; Gao, S.; McKenzie, G.; Hu, Y.; Bhaduri, B. GeoAI: Spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. *Int. J. Geogr. Inf. Sci.* **2020**, *34*, 625–636. [CrossRef]
2. Liu, X.; Kounadi, O.; Zurita-Milla, R. Incorporating spatial autocorrelation in machine learning models using spatial lag and eigenvector spatial filtering features. *ISPRS Int. J. Geo-Inf.* **2022**, *11*, 242. [CrossRef]
3. Nikparvar, B.; Thill, J.C. Machine learning of spatial data. *ISPRS Int. J. Geo-Inf.* **2021**, *10*, 600. [CrossRef]
4. Li, W. GeoAI: Where machine learning and big data converge in GIScience. *J. Spat. Inf. Sci.* **2020**, *20*, 71–77. [CrossRef]
5. Li, W.; Hsu, C.Y.; Hu, M. Tobler's First Law in GeoAI: A spatially explicit deep learning model for terrain feature detection under weak supervision. *Ann. Am. Assoc. Geogr.* **2021**, *111*, 1887–1905. [CrossRef]
6. Credit, K. Spatial models or random forest? Evaluating the use of spatially explicit machine learning methods to predict employment density around new transit stations in Los Angeles. *Geogr. Anal.* **2022**, *54*, 58–83. [CrossRef]
7. Hengl, T.; Nussbaum, M.; Wright, M.N.; Heuvelink, G.B.M.; Gräler, B. Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables. *PeerJ* **2018**, *6*, e5518. [CrossRef]
8. Das, S.S.S.; Ali, M.E.; Li, Y.F.; Kang, Y.B.; Sellis, T. Boosting house price predictions using geo-spatial network embedding. *Data Min. Knowl. Discov.* **2021**, *35*, 2221–2250. [CrossRef]
9. Griffith, D.A. What is spatial autocorrelation? Reflections on the past 25 years of spatial statistics. *Espace Géogr.* **1992**, *21*, 265–280. [CrossRef]
10. Emmanuel, T.; Maupong, T.; Mpoeleng, D.; Semong, T.; Mphago, B.; Tabona, O. A survey on missing data in machine learning. *J. Big Data* **2021**, *8*, 140. [CrossRef]
11. Hastie, T.; Tibshirani, R.; James, G.; Witten, D. *An Introduction to Statistical Learning*, 2nd ed.; Springer: New York, NY, USA, 2021; Chapter 4, pp. 160–163.
12. Griffith, D.A.; Agarwal, K.; Chen, M.; Lee, C.; Panetti, E.A.; Rhyu, K.; Venigalla, L.; Yu, X. Geospatial socio-economic/demographic data: The existence of spatial autocorrelation mixtures in georeferenced data—Part I. *Trans. GIS* **2022**, *26*, 72–87.
13. Griffith, D.A. Spatial autocorrelation and art. *Cybergeo* **2016**. Available online: http://journals.openedition.org/cybergeo/27429 (accessed on 16 January 2026).
14. Li, T.; Shen, H.; Yuan, Q.; Zhang, X.; Zhang, L. Estimating ground-level PM2.5 by fusing satellite and station observations: A geo-intelligent deep learning approach. *Geophys. Res. Lett.* **2017**, *44*, 11985–11993. [CrossRef]
15. Kiely, T.J.; Bastian, N.D. The spatially conscious machine learning model. *Stat. Anal. Data Min.* **2020**, *13*, 31–49. [CrossRef]
16. Behrens, T.; Schmidt, K.; Viscarra Rossel, R.A.; Gries, P.; Scholten, T.; MacMillan, R.A. Spatial modelling with Euclidean distance fields and machine learning. *Eur. J. Soil Sci.* **2018**, *69*, 757–770. [CrossRef]
17. Chen, L.; Ren, C.; Li, L.; Wang, Y.; Zhang, B.; Wang, Z.; Li, L. A comparative assessment of geostatistical, machine learning, and hybrid approaches for mapping topsoil organic carbon content. *ISPRS Int. J. Geo-Inf.* **2019**, *8*, 174. [CrossRef]
18. Foresti, L.; Pozdnoukhov, A.; Tuia, D.; Kanevski, M. Extreme precipitation modelling using geostatistics and machine learning algorithms. In *geoENV VII–Geostatistics for Environmental Applications*; Springer: Berlin, Germany, 2010; pp. 41–52.
19. Georganos, S.; Grippa, T.; Niang Gadiaga, A.; Linard, C.; Lennert, M.; Vanhuysse, S.; Mboga, N.; Wolff, E.; Kalogirou, S. Geographical random forests: A spatial extension of the random forest algorithm to address spatial heterogeneity in remote sensing and population modelling. *Geocarto Int.* **2021**, *36*, 121–136. [CrossRef]
20. Paliwal, M.; Kumar, U.A. Neural networks and statistical techniques: A review of applications. *Expert Syst. Appl.* **2009**, *36*, 2–17. [CrossRef]
21. Baldominos, A.; Blanco, I.; Moreno, A.J.; Iturrarte, R.; Bernárdez, Ó.; Afonso, C. Identifying real estate opportunities using machine learning. *Appl. Sci.* **2018**, *8*, 2321. [CrossRef]
22. Hagenauer, J.; Omrani, H.; Helbich, M. Assessing the performance of 38 machine learning models: The case of land consumption rates in Bavaria, Germany. *Int. J. Geogr. Inf. Sci.* **2019**, *33*, 1399–1419. [CrossRef]
23. Sanjar, K.; Bekhzod, O.; Kim, J.; Paul, A.; Kim, J. Missing data imputation for geolocation-based price prediction using KNN-MCF method. *ISPRS Int. J. Geo-Inf.* **2020**, *9*, 227. [CrossRef]
24. Dudani, S.A. The Distance-Weighted k-Nearest-Neighbor Rule. *IEEE Trans. Syst. Man Cybern.* **1976**, *6*, 325–327. [CrossRef]

25. Atkinson, P.M. Spatially weighted supervised classification for remote sensing. *Int. J. Appl. Earth Obs. Geoinf.* **2004**, *5*, 277–291. [CrossRef]
26. Shen, X.; Lin, Z.; Brandt, J.; Avidan, S.; Wu, Y. Object retrieval and localization with spatially-constrained similarity measure and k-NN re-ranking. In Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Providence, RI, USA, 16–21 June 2012; pp. 3013–3020.
27. Mohd, T.; Jamil, N.S.; Johari, N.; Abdullah, L.; Masrom, S. An overview of real estate modelling techniques for house price prediction. In *Charting a Sustainable Future of ASEAN in Business and Social Sciences*; Springer: Singapore, 2020; pp. 321–338.
28. IBM. What Is the K-Nearest Neighbors Algorithm? Available online: https://www.ibm.com/think/topics/knn (accessed on 20 November 2025).
29. Meyer, H.; Pebesma, E. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods Ecol. Evol.* **2021**, *12*, 1620–1633. [CrossRef]
30. Peterson, L.E. K-Nearest Neighbor. *Scholarpedia* **2009**, *4*, 1883. [CrossRef]
31. Razavi-Termeh, S.V.; Sadeghi-Niaraki, A.; Razavi, S.; Choi, S.M. Enhancing flood-prone area mapping: Fine-tuning the K-nearest neighbors (KNN) algorithm for spatial modelling. *Int. J. Digit. Earth* **2024**, *17*, 2311325. [CrossRef]
32. Koldasbayeva, D.; Tregubova, P.; Gasanov, M.; Zaytsev, A.; Petrovskaia, A.; Burnaev, E. Challenges in data-driven geospatial modeling for environmental research and practice. *Nat. Commun.* **2024**, *15*, 10700. [CrossRef] [PubMed]
33. Zhang, L. Housing price prediction using machine learning algorithm. *J. World Econ.* **2023**, *2*, 18–26. [CrossRef]
34. Zulkifley, N.H.; Abdul Rahman, S.; Ubaidullah, N.H.; Ibrahim, I. House price prediction using a machine learning model: A survey of literature. *Int. J. Mod. Educ. Comput. Sci.* **2020**, *12*, 46–54. [CrossRef]
35. Rico-Juan, J.R.; Taltavull de La Paz, P. Machine learning with explainability or spatial hedonics tools? An analysis of asking prices in Alicante, Spain. *Expert Syst. Appl.* **2021**, *171*, 114590. [CrossRef]
36. Gamper-Rabindran, S.; Timmins, C. Does cleanup of hazardous waste sites raise housing values? Evidence of spatially localized benefits. *J. Environ. Econ. Manag.* **2013**, *65*, 345–360. [CrossRef]
37. Joseph, V.R. Optimal ratio for data splitting. *Stat. Anal. Data Min.* **2022**, *15*, 531–538. [CrossRef]
38. Ploton, P.; Mortier, F.; Réjou-Méchain, M.; Barbier, N.; Picard, N.; Rossi, V.; Dormann, C.; Cornu, G.; Viennois, G.; Bayol, N.; et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nat. Commun.* **2020**, *11*, 4540. [CrossRef]
39. Salazar, J.J.; Garland, L.; Ochoa, J.; Pyrcz, M.J. Fair train-test split in machine learning: Mitigating spatial autocorrelation for improved prediction accuracy. *J. Pet. Sci. Eng.* **2022**, *209*, 109885. [CrossRef]
40. Meyer, H.; Pebesma, E. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nat. Commun.* **2022**, *13*, 2208. [CrossRef] [PubMed]
41. Jiao, Z.; Tao, R. Geographical Gaussian process regression: A spatial machine-learning model based on spatial similarity. *Geogr. Anal.* **2025**, *57*, 507–520. [CrossRef]
42. Limsombunc, V.; Gan, C.; Lee, M. House price prediction: Hedonic price model vs. artificial neural network. *Am. J. Appl. Sci.* **2004**, *1*, 193–201. [CrossRef]
43. Yazdani, M. Machine learning, deep learning, and hedonic methods for real estate price prediction. *arXiv* **2021**, arXiv:2110.07151. [CrossRef]
44. Kopczewska, K. Spatial machine learning: New opportunities for regional science. *Ann. Reg. Sci.* **2022**, *68*, 713–755. [CrossRef]
45. Bellman, R.E. *Dynamic Programming*, 6th Printing; Princeton University Press: Princeton, NJ, USA, 1957.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.